



Influence of Anchor Prestress Distribution on the Response of an Anchored Deep Excavation-Soil-Building System under Seismic Loading

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Abstract: Deep excavations constructed in densely built urban environments are frequently supported by prestressed anchor systems, whose performance can significantly influence the stability of adjacent soil and structures. In this study, a finite element model was developed in PLAXIS 2D to investigate the mechanical response of an anchored deep excavation subjected to both static and seismic conditions. The excavation was retained by diaphragm walls and supported by three levels of prestressed anchors, while a five-story building was located at a distance of 10 m from the excavation boundary. Four representative prestress distribution patterns were considered, including uniform, decreasing, increasing, and mid-level maximum distributions. The influence of these prestressing schemes was evaluated through a comprehensive analysis of anchor forces, diaphragm wall lateral displacements, building horizontal displacements, and foundation settlements. The results indicated that the initial prestress distribution exerted a pronounced influence on the system behavior under static conditions. Reductions in wall deformation and building settlement were found to be strongly dependent on the magnitude and distribution of the initial prestressing forces. Among the investigated strategies, the decreasing prestressing scheme provided the most favorable overall performance. In contrast, the mid-level maximum distribution scheme exhibited comparatively lower efficiency in limiting excavation-induced deformations. Under seismic loading, however, the differences among the prestress distribution patterns became negligible. These findings suggest that optimization of anchor prestress distribution is primarily beneficial for controlling excavation-induced deformation under static conditions, whereas its effectiveness becomes considerably less pronounced when seismic effects dominate system behavior. The study provides practical guidance for the design and optimization of prestressed anchoring systems for deep excavations located in seismically active urban areas.

Keywords: Deep excavation; Anchor prestressing; Finite element modeling; Pit wall; PLAXIS 2D

1 Introduction

In the modern world, the increasing number of buildings constructed every year is associated with global population growth and urbanization. Various business centers, multi-story residential buildings, and skyscrapers are being built in close proximity to one another. Construction work near an existing building has a significant impact on the behavior of the entire surrounding environment. The foundations of buildings located close to one another are subject to mutual influence. Poulos [1] and Van Nguyen et al. [2] studied this process to identify the main factors affecting building foundations. It was found that soil parameters can significantly affect building settlement. For example, foundations constructed in soft soils may experience greater settlement, which can induce additional settlement in adjacent structures through soil-structure interaction effects. Similarly, a structure supported by a deeper foundation system may influence the stress state of the surrounding soil, thereby increasing the settlement of an adjacent structure founded at a shallower depth. The current study examines a deep excavation pit adjacent to an existing five-story building on a pile foundation. The primary objective is to study the impact of anchor prestress distribution on the excavation-soil-building system using a numerical model developed in PLAXIS 2D.

The impact of anchor systems of a deep excavation pit on the soil and surrounding environment has been studied for a long time. Maleki et al. [3] comprehensively examined the enclosing walls of a deep excavation pit using anchor

systems. By considering a large number of numerical models with varying excavation depths, soil characteristics, wall stiffness, anchor parameters, and their inclination angle, it was found that the primary criterion in the design of walls supported by anchors is the horizontal displacement of the excavation wall. The maximum displacement of the excavation pit wall was proposed to be limited to $0.002H$, where H is the excavation depth. Furthermore, it was noted that increasing the length and number of anchors leads to a reduction in deformation. However, after reaching a certain value, further increases in these parameters become economically unviable. Pratama et al. [4] investigated the effect of anchor systems on the stability of the enclosing structures of a deep excavation. The analysis was conducted using a numerical model in the PLAXIS 2D program. The excavation walls were constructed using secant piles. The main objective of the study was to determine the optimal anchor parameters for the effective operation of secant piles in the excavation. The smallest horizontal displacements were observed at zero degrees, i.e., with horizontal anchors. However, under these conditions, it is necessary to increase the anchor length. Subsequently, it was concluded that using an oblique anchor angle of 45° is the most effective, as the tie lengths of such anchors should not be maximized. Furthermore, it was found that increasing the number of anchors beyond two does not improve wall displacements. In turn, increasing the prestress of the anchors is effective only up to a certain value (up to 200 kN). Further increase in prestress leads to an increase in the bending moment of the intersecting piles.

In addition, Zhussupbekov et al. [5] examined the construction of a diaphragm wall and its anchors during the excavation of a deep pit under challenging geological conditions in Astana, Kazakhstan. The main stages of the diaphragm wall construction were described in detail. The process began with the installation of a guide wall, which secured the position of the future trench. After this stage, the trench for the future diaphragm wall was dug. Bentonite mortar was used to support the trench, creating pressure on the trench walls and preventing its collapse. This trench was then reinforced and filled with concrete from the bottom up, thereby displacing the bentonite. The process of installing diaphragm wall anchors was also described in this study. Anchors were installed by drilling holes through the diaphragm wall into the soil, after which the anchors were installed and secured with cement mortar. Anchors are known to help reduce deformations in diaphragm walls.

The impact of lowering the groundwater level on buildings located near a deep excavation pit was investigated by some studies [6, 7]. Building settlement increases due to soil compaction. Furthermore, not all building parts settle uniformly, which increases differential settlement. Soil particles are washed out with the water, forming voids and reducing soil rigidity. The impact of a deep excavation pit on the surrounding environment and adjacent buildings was assessed by Tomczak and Mitew-Czajewska [8]. It is worth noting that the study site is located in central Warsaw, Poland, and is surrounded by tenement houses on three sides, with a metro line on the fourth side. The analysis was conducted based on both a numerical model and in-situ data. The results of the numerical modeling showed that the settlement values of the adjacent buildings significantly exceeded the permissible values without additional soil stabilization measures. Therefore, it was decided to strengthen the foundations using jet grouting under the most heavily loaded building foundations. A repeated analysis of the building settlements showed a significant reduction to acceptable values. Furthermore, the results of the numerical modeling and in-situ data were compared after strengthening the building foundations. The discrepancies in displacement values were insignificant, demonstrating the high accuracy of the calculations.

Chen et al. [9] employed three-dimensional coupled consolidation finite element model to study the effects of diaphragm wall construction in saturated soft clay. It was found that pore water pressure, soil arching, and permeability have significant effect on stress changes and ground movements.

Furthermore, Al-Ne'aimi and Nasir [10] identified the optimal number of anchor levels. It was found that four anchor levels can significantly improve the horizontal displacements of the excavation wall, while further increasing the number of anchors has virtually no effect. In addition, it was found that the lower the groundwater level, the smaller the horizontal displacements, wall bending moments, and soil settlements.

Maleki and Pak [11] developed a new multivariate adaptive regression splines method for designing anchor systems for deep excavation enclosure structures. Based on existing finite element models, the relationships for estimating the maximum horizontal displacements of diaphragm excavation walls were developed. The suitability of this method was confirmed by comparing the model results with field observation data. The model demonstrated good agreement with measured values of real-world data, demonstrating the model's high accuracy. Furthermore, it was found that the excavation height, soil cohesion, and its internal friction angle have the greatest influence on horizontal wall displacements. The importance of these parameters was also confirmed by Jasmine Nisha and Muttharam [12], who compared the performance of a numerical model of a deep excavation with field data. It was found that the finite element method can accurately predict the behavior of excavation walls with the correct input of soil parameters. Ünver and Ünver [13] applied a similar data comparison approach to studying the enclosing structures of a deep excavation made of bored piles with anchors. A reverse analysis was performed, comparing and refining the numerical model data based on field data. This approach also confirmed that the horizontal displacements of the excavation walls depend on the deformation characteristics of the soil.

Jalili et al. [14] examined the anchor system of the enclosing structures of a deep excavation under seismic

environmental conditions. The main objective of this work was to determine the optimal distribution of anchors that would reduce the horizontal displacements of the excavation wall without changing the length of the anchors. It was found that the height distribution of anchors significantly affects the horizontal displacements of the wall. The lower rows of anchors are primarily responsible for ensuring overall stability, while the upper anchors limit deformations in the upper zone of the wall. A diamond distribution was used, in which the highest anchor density occurs in zones of maximum bending moments, improving the system's efficiency and reducing deformations. The performance of diaphragm walls was evaluated through numerical and experimental studies by Lentini and Castelli [15]. A distinctive feature of this work was the use of inclinometers and accelerometers on a section of the wall, which subsequently allowed for obtaining data on the wall's behavior during construction. As in other similar studies, the data obtained from the section of the wall were compared with a numerical model in the PLAXIS program. The results of the study showed that the numerical model accurately predicts the horizontal displacements of the wall, demonstrating its applicability in assessing the deformation state of enclosing structures.

Unlike other studies focused on anchor parameters, Saadi et al. [16] examined the effect of diaphragm wall geometry on the entire soil-excavation system under seismic environmental conditions. A zigzag arrangement of diaphragm wall panels was proposed, as this, according to calculations, reduces deformations under the dynamic impact of an earthquake. Furthermore, this wall geometry does not require an increase in wall thickness, but it redistributes stresses and eliminates stress concentrations at wall joints. Maleki and Nabizadeh [17] studied the effect of seismic action on a deep excavation pit with a guardian truss structures system. A comparison of key parameters, such as wall displacement, settlement, and soil heaving, revealed that seismic action worsens the condition of the entire system, increasing them all. Furthermore, reducing the number of struts weakens the system. Improving key soil parameters (strain modulus, cohesion, and friction angle) significantly reduces system deformations.

Zhao et al. [18] investigated the impact of anchor failure on pile walls in a deep excavation. It was found that anchor failure directly impacts the performance of the excavation's enclosing structures, as it affects wall deformations and pit stability. It's worth noting that the failure of a single anchor has little effect on wall deformations, but the failure of a single row of anchors has a moderate impact. Failure of two or more rows leads to a significant deterioration in the system's condition. Furthermore, it was found that anchor failure leads to additional loads on adjacent anchors and an increase in their prestress. A chain reaction is also possible, in which the failure of one anchor leads to the failure of the adjacent anchor. When the upper anchors fail, the top of the wall suddenly becomes unsupported, leading to a sharp increase in displacements and bending moments of the excavation wall. Hsu et al. [19] and Farghaly [20] studied the effect of a deep excavation pit on adjacent buildings. It was found that the distance between the pit and the building is a key factor influencing the settlement and deformations of both the pit and the building. In addition, the presence of a basement can also affect building deformations. Low basement walls are more susceptible to soil deformations because they are located in higher soil layers. In seismic environments, it is important to pay attention to the earthquake direction. A building may deform more if the source of vibrations first passes through a pit, i.e., a part of the soil with reduced rigidity. Dmochowski and Szolomicki [21] proposed a design approach for deep excavations that facilitated prompt response when permissible wall displacements were exceeded. Godlewski et al. [22] described several risk groups during the construction of underground structures. Soil parameters, groundwater level, and soil aggressiveness were included in one group. Building characteristics such as year of construction, foundation type, and deterioration constituted a second risk group. Potential consequences during construction work were included in a third group.

From a sustainability perspective, the selection of an appropriate anchor prestressing scheme is important because it may improve the safety and integrity of urban excavation systems. Proper prestressing design has the potential to reduce wall deformation, ground settlement, and minimize the risk of damage to adjacent buildings and infrastructure, particularly in densely populated cities and under seismic conditions. Therefore, understanding the influence of anchor prestressing on excavation pit behavior is essential for developing more sustainable, resilient, and cost-effective urban infrastructure systems.

2 Methodology

To study the effect of the prestressing scheme on the pit-soil-building system, a numerical model of a deep excavation adjacent to an existing building on a pile foundation was constructed in PLAXIS 2D. The geometric dimensions and enclosing walls of the excavation pit, the distance between the building and the pit, and the dimensions of the building and its foundation were kept constant to assess the impact of the anchor prestressing scheme on the system. Figure 1 shows the initial computational model and its main geometric parameters as well as the boundary conditions.

2.1 Characteristics of Soil, the Building, and Excavation Walls

The engineering and geological conditions were defined as three layers with different physical and mechanical properties. For accurate analysis of the deformation state of the soil, excavation walls, and the building, the hardening

soil model was used. Unlike other models, this model allows for a more accurate description of the loading and unloading processes of the soil during different construction stages.

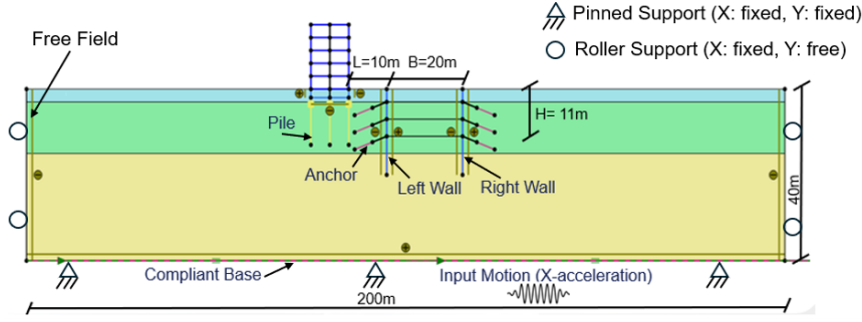


Figure 1. Finite element model of the anchored excavation-soil-building system

The upper sand layer is 3 m thick. This layer rests on a lower sand layer of medium-grain size, which is 12 m thick. The bottommost layer is 25 m thick, consisting of loam. All physical and mechanical properties of the soil layers are presented in Table 1.

Table 1. Physical and mechanical parameters of soil layers

Name	Symbol	Upper Sand	Lower Sand	Loam	Unit of Measurement
Secant stiffness in triaxial loading	E_{50}^{ref}	35000	30000	12000	kN/m ²
Oedometer stiffness	$E_{\text{oed}}^{\text{ref}}$	35000	30000	8000	kN/m ²
Unloading-reloading stiffness	$E_{\text{ur}}^{\text{ref}}$	94840	90000	36000	kN/m ²
Stiffness exponent	m	0.5	0.5	0.8	—
Friction angle	ϕ	32	34	29	°
Dilatancy angle	ψ	0	4	0	°
Unsaturated unit weight	γ_{unsat}	19	20	17	kN/m ³
Saturated unit weight	γ_{sat}	20	20	19	kN/m ³

Note: The superscript “ref” denotes the reference stiffness parameters used in the PLAXIS Hardening Soil model. The dash (—) indicates that the unit is not applicable because the parameter is dimensionless.

The excavation pit walls were represented in the model as linear elastic elements with a specified unit weight of $w = 8.3 \text{ kN/m}^2$. The walls were modeled as diaphragm walls made of reinforced concrete with a Young’s modulus of $E = 34.64 \text{ GPa}$ and a thickness of $t \approx 0.35 \text{ m}$. The excavation pit walls were embedded in the ground to a depth of 20 m, providing sufficient resistance to soil deformations. The diaphragm walls are one of the main elements that support lateral soil pressure and loads from each anchor level. Horizontal wall movements affect not only changes in the soil but also the behavior of the entire system, including the building and its foundation. It is worth noting that the parameters of the excavation pit walls were constant in all simulation models, as was the distance between the building and the excavation pit, which was 10 m.

The building located next to the excavation pit has five floors, each 3 m high, with a 2-m basement. It is presented in the form of “plate” elements and rests on a pile foundation. To improve the building’s stability, the system includes columns in the form of “anchor” elements. They transfer the vertical load from the building to the foundation. The building’s foundation consists of three rows of piles, 11 m long. It rests on a second soil layer of medium-grain sand. The building’s piles are made of reinforced concrete with a Young’s modulus of $32.84 \times 10^6 \text{ kN/m}^2$ and a pile diameter of 1 m. This choice of foundation type is explained by the fact that pile foundations effectively resist soil deformations in close proximity to a new, deep excavation pit.

2.2 Anchors and the Prestressing Scheme

To reduce the horizontal displacement of the excavation pit walls, the numerical model included three levels of anchors. The upper level is located at a depth of 3 m from the surface, the middle level is at 7 m, and the lower level is at 11 m. The anchors were designed as linear elastic elements consisting of free and loaded parts. The free length of the anchors in the model was defined as an “anchor” element with an axial anchor stiffness of $EA = 500 \times 10^3 \text{ kN}$. The bond length was represented in the model as an “embedded beam” element with a Young’s modulus of $7.07 \times 10^6 \text{ kN/m}^2$ and a diameter of $d = 0.3 \text{ m}$. The anchor inclination angle was assumed to be the same in all models, equal to 20°.

The bond length of the anchors was designed using Eq. (1) [23]:

$$\text{Bond length} = \frac{T \times SF}{\text{Estimated ultimate transfer load}} \quad (1)$$

where, T and SF denote the anchor load and the safety factor, respectively, with SF generally taken as 2. In this paper, bond length = $\frac{400 \text{ kN} \times 2}{190 \text{ kN/m}} = 4.2 \text{ m}$.

The anchor load T was taken as the average of all anchor prestress values. Prestress data are presented in Table 2. The estimated ultimate transfer load [23] was adopted for dense sand, where the pit wall anchors are located. Given that the bond length is typically taken to be between 4.5 and 12 m, in this case, it is appropriate to adopt a loaded length of 4.5 m. Furthermore, it is suggested that the minimum free length of a wire anchor is 4.5 m, and for a cable anchor, 3 m. Therefore, the free length of the anchors was adopted as 4.5 m.

Table 2. Parameters of the anchor systems for different prestress loading schemes

Anchor No.	Installation Depth (m)	Scheme 1	Scheme 2	Scheme 3	Scheme 4
		Uniform	Decreasing	Increasing	Maximum in the Middle
1	3	400	460	340	360
2	7	400	400	400	480
3	11	400	340	460	360

This study investigated the effect of anchor prestress loading schemes on the behavior of the pit walls and the adjacent building. For this purpose, four different anchor prestress loading schemes [23] were adopted.

It is worth noting that the total length of an anchor is 9 m, the inclination angle is 20°, and the average prestress value is 400 kN.

2.2.1 Numerical modeling assumptions

To summarize the assumptions and settings adopted in numerical modeling, a detailed description is provided in Table 3. As shown in the table, the seismic analysis of geotechnical earthquake engineering problems in PLAXIS 2D consists of two phases: static and dynamic [24–26]. During the static phase, the Mohr-Coulomb constitutive model was employed because the analysis was limited to establishing the initial stress state. No external seismic loading was applied in this phase. Instead, a plastic analysis was performed to establish the initial stress state and allow the model to reach equilibrium. The assumptions and modeling parameters considered in this phase are summarized in Table 3. In the dynamic phase, where the nonlinear behavior of the soil was considered, the more advanced Hardening Soil constitutive model was adopted. In this phase, the selected earthquake record was applied at the base of the model, while the boundary conditions and damping parameters were defined according to the values presented in Table 3.

Table 3. Summary of assumptions for the numerical modeling

Modeling Phase	Structure	Element (Material)	Boundary Conditions	Mesh Size or Mesh Refinement of the Soil Layer	Damping Assumptions
Static	Building	Plate (elastic)	Lateral boundaries: Roller (x-direction: fixed; y-direction: free) Bottom boundary: Pinned (x-direction: fixed; y-direction: fixed)	Meshing with triangular shapes was used. Mesh size in the model was fine.	–
	Anchor	Node-to-node anchor (elastic)			
	Pile	Embedded-beam row (elastic)			
	Soil	Mohr-Coulomb			
Dynamic	Building	Plate (elastic)	Lateral boundaries: Roller (x-direction: fixed; y-direction: free) Bottom boundary: Pinned (x-direction: fixed; y-direction: fixed)	Meshing with triangular shapes was used. Mesh size in the model was fine.	Building: Damping = 5% $(\alpha = 0.57, \beta = 0.014)$ Sandy layer: Damping = 10% $(\alpha = 0.11, \beta = 0.029)$ Loam layer: Damping = 15% $(\alpha = 0.17, \beta = 0.043)$
	Anchor	Node-to-node anchor (elastic)			
	Pile	Embedded-beam row (elastic)			
	Soil	Hardening soil			

Note: The dash (–) indicates that the corresponding damping assumptions are not applicable for the static analysis.

2.3 Seismic Impact

The present study aimed to examine the behavior of the excavation walls and the adjacent building under static and seismic conditions. To study the effect of seismic impact on the anchored excavation-soil-building system, a horizontal acceleration record from the 1994 Northridge earthquake, recorded at the Castaic-Old Ridge Route station, was applied at the base of the models. The peak ground acceleration of this record is 0.55 g. The reason for selecting this earthquake is its high magnitude, which allows assessment of the effects of a strong seismic event on the simulated models. The time history of the acceleration is presented in Figure 2.

Figure 2 shows that the earthquake intensity reaches its maximum value within the first 10 seconds. To study the system's behavior during an earthquake, a comparative analysis was conducted in terms of the horizontal displacements of the excavation pit and building walls, anchor forces, and building settlement in the static and seismic states of the system.

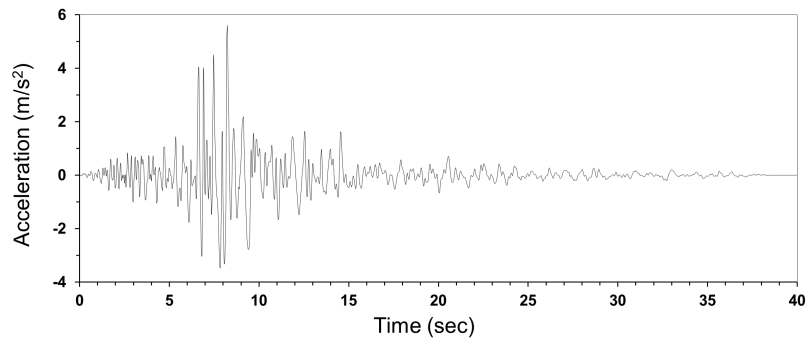


Figure 2. Acceleration time history of the 1994 Northridge earthquake

3 Results

The key indicators for studying the effect of the prestressing scheme on the excavation-soil-building system are changes in the longitudinal anchor forces. Table 4 presents the anchor forces in the static and seismic states. Graphs of changes in anchor forces were constructed in accordance with the table. Figure 3 shows that, in the static state of the system, the initial prestress patterns determine the behavior of each anchor level. All patterns differ in the distribution of forces across depth: uniform, decreasing, increasing, and mid-level maximum distributions. However, after a seismic event, the forces increase at each level. Regardless of the initial prestress pattern, the curve shapes become similar.

Table 4. Axial forces in anchors for different prestress loading schemes under static and seismic conditions

Anchor No.	Depth/Side	Scheme 1		Scheme 2		Scheme 3		Scheme 4	
		Static	Seismic	Static	Seismic	Static	Seismic	Static	Seismic
1	3 m/close to the building	399.2	551.9	459.1	571.1	339.2	532.8	359.3	532.3
2	7 m/close to the building	398.9	757.1	398.8	761	398.9	750.3	478.6	799
3	11 m/close to the building	399.5	809.2	339.6	782.8	459.4	830.9	359.6	795.1
4	3 m/far from the building	399	545.3	458.7	574.1	339.1	522.6	359.1	519.2
5	7 m/far from the building	398.7	715.6	398.7	720.1	398.7	709	478.7	762.1
6	11 m/far from the building	399.5	841.3	339.6	808.6	459.4	865	359.6	825.2

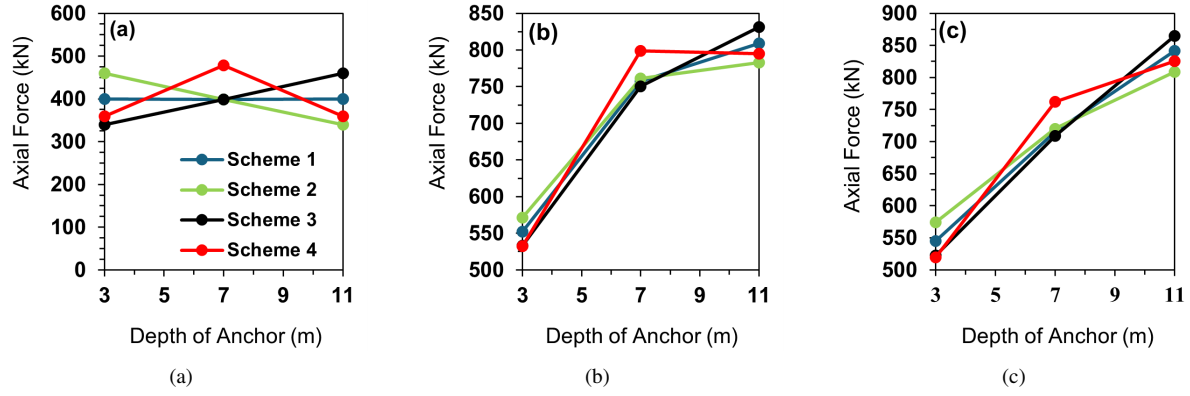


Figure 3. Axial forces in anchors: (a) before the earthquake; (b) after the earthquake in the left wall; and (c) after the earthquake in the right wall

In Scheme 1, a uniform distribution of forces is observed with increasing depth. In Scheme 2, the highest prestress is observed at the top anchor, and the values of anchor prestress are decreasing along the pit wall depth. Scheme 3 exhibits an increasing force distribution with depth. The curve in Scheme 4 differs slightly from the others, but this is explained by the fact that the middle anchor level initially has higher prestress than the other anchors. Furthermore, the results show that the building influences the distribution of forces after an earthquake. It preloads the soil, thereby compacting it and reducing soil deformation. The building and its foundation restrict soil movement toward the excavation pit, thereby reducing the forces in the anchors compared to the side without the building. On the opposite side, the soil deforms more freely, creating greater lateral pressure on the excavation pit wall.

To quantitatively assess the distribution of anchor forces on different sides of the pit, the non-uniformity coefficient k was calculated using Eq. (2). Data for all prestressing schemes are presented in Table 5.

$$k = \frac{N_{\max}}{N_{\text{avg}}} \quad (2)$$

Table 5. Force distribution non-uniformity coefficient (k) values in different anchor prestressing schemes

Scheme	Side of the Pit Wall	k	
		Static	Seismic
Scheme 1	Close to the building	1.001	1.146
	Far from the building	1.001	1.201
Scheme 2	Close to the building	1.150	1.110
	Far from the building	1.150	1.154
Scheme 3	Close to the building	1.151	1.179
	Far from the building	1.151	1.238
Scheme 4	Close to the building	1.199	1.127
	Far from the building	1.199	1.175

Table 5 shows that, in a static state, Scheme 1 is the most uniform. Anchors at all levels in this scheme have a prestress of 400 kN, resulting in a non-uniformity coefficient of 1. Scheme 4, in a static state, exhibits the highest non-uniformity coefficient, demonstrating a pronounced non-uniformity in the distribution of anchor forces. This is confirmed by Figure 3a, which shows the highest force at the middle anchor level. After the earthquake, the minimum non-uniformity coefficient ($k = 1.11$) for the wall anchors on the building side is observed in the descending scheme in Scheme 2, while the maximum is 1.179 in Scheme 3. The difference between the values is small, but the descending scheme still results in a more uniform distribution of anchor forces.

In Scheme 2, the upper anchors have the maximum initial prestress. After the onset of seismic action, they begin to work and limit deformations in the upper part of the wall. The lower anchors, however, begin to work smoothly, since some of the deformations are limited by the upper anchors. On the opposite side, where the building is absent, the non-uniformity coefficient values are higher: the maximum ($k = 1.238$) is observed for Scheme 3, and the minimum ($k = 1.154$) is for Scheme 2. This is due to the greater soil pliability in the absence of the building, which leads to uneven soil deformations.

3.1 Horizontal Displacements of Excavation Pit Walls

Horizontal displacements of the walls are an important indicator of the performance of excavation enclosing structures. Unlike forces, which demonstrate the internal work of anchors, horizontal displacements indicate the deformed behavior of the wall under static and seismic conditions. Therefore, a comparative analysis of horizontal displacements of walls under different prestressing schemes was performed. Figure 4 shows the horizontal displacements of the excavation pit walls in the static state of the system.

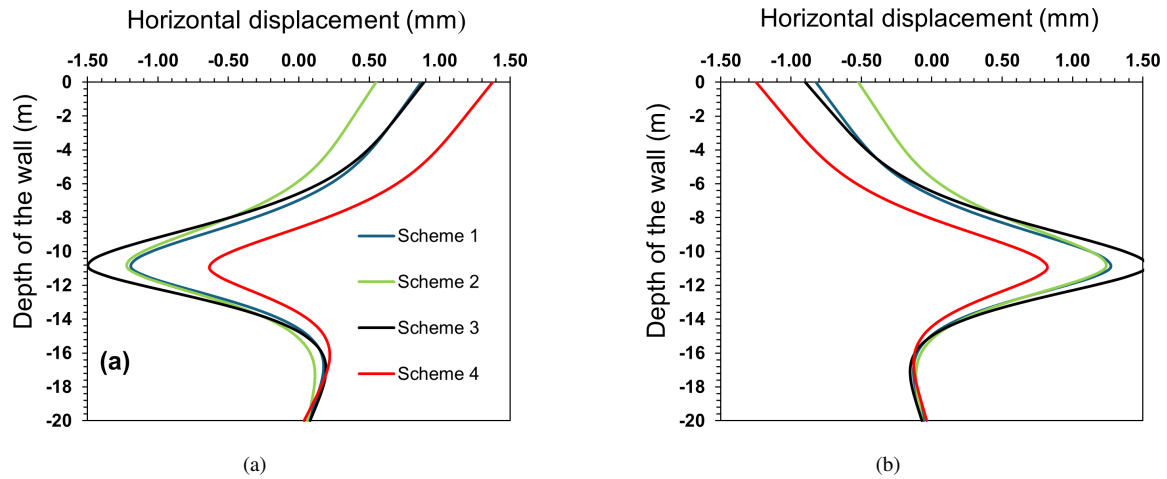


Figure 4. Horizontal displacements of the pit wall before the earthquake: (a) left wall; (b) right wall

As shown in the figure, the deformations of the excavation pit wall before the earthquake are directly dependent on the initial prestress values at each anchor level. The minimal displacement of the upper part of the wall on the building side is observed in the decreasing pattern in Scheme 2 (approximately 0.5 mm), as the initial prestress in the upper part is maximum (460 kN). With the increasing pattern in Scheme 3, the maximum wall deflection is observed in the middle compared to other patterns. This is because, with this scheme, the bottom anchor is the most tensioned, sharply limiting wall displacement in this area. The position of maximum deflection shifts toward the middle of the wall, resulting in maximum displacement in this area.

Furthermore, Figure 4 shows that the displacement of the upper part of the walls on the building side is greater than that on the opposite side. The building acts as an additional load on the soil, causing the wall on the building side to be initially loaded more heavily. Subsequently, during excavation, the wall shifts more toward the excavation pit. In the middle, everything changes. The horizontal displacements are greater on the side where the building is not present. At this depth, the walls on the building side are constrained by the foundation, which limits horizontal displacement. On the opposite side, the soil is looser and less rigid due to the absence of the building foundation. Therefore, the walls under these conditions can develop greater curvature.

To assess the seismic impact on the horizontal displacements of the excavation walls, graphs were constructed to reflect the behavior of the walls during and after the earthquake, as shown in Figure 5 and Figure 6. An earthquake significantly deforms the walls of a deep pit. Figure 5 shows that the main deformations occur during the earthquake's peak intensity. Horizontal displacements increase several times. Furthermore, it is clear that the top and bottom of the walls behave differently. The top of the wall is the most vulnerable due to the lack of pressure from other soil layers above. This part deforms very easily during an earthquake. The lower part of the wall is deeply embedded in the ground, and therefore, horizontal displacements in this part are much smaller than those in the upper part. Soil pressure limits soil movement during an earthquake.

The walls on the building side and the free side change similarly, but if comparing their values, it can be seen that horizontal displacements are slightly greater for the walls on the building side. This is because during an earthquake, the ground and the building itself begin to vibrate. These vibrations create acceleration in the building and inertial forces within it, which are transmitted through the building's foundation into the ground. The walls of the pit are subjected to additional load and experience greater displacement. After an earthquake, the horizontal displacements of the pit walls remain at the same level as during the earthquake. After the earthquake, the curves for the different anchor prestressing schemes are virtually identical. This indicates that under seismic loading, the effect of the initial prestressing becomes insignificant. The main factors are soil deformation and its lateral pressure on the excavation pit walls, resulting in wall displacement.

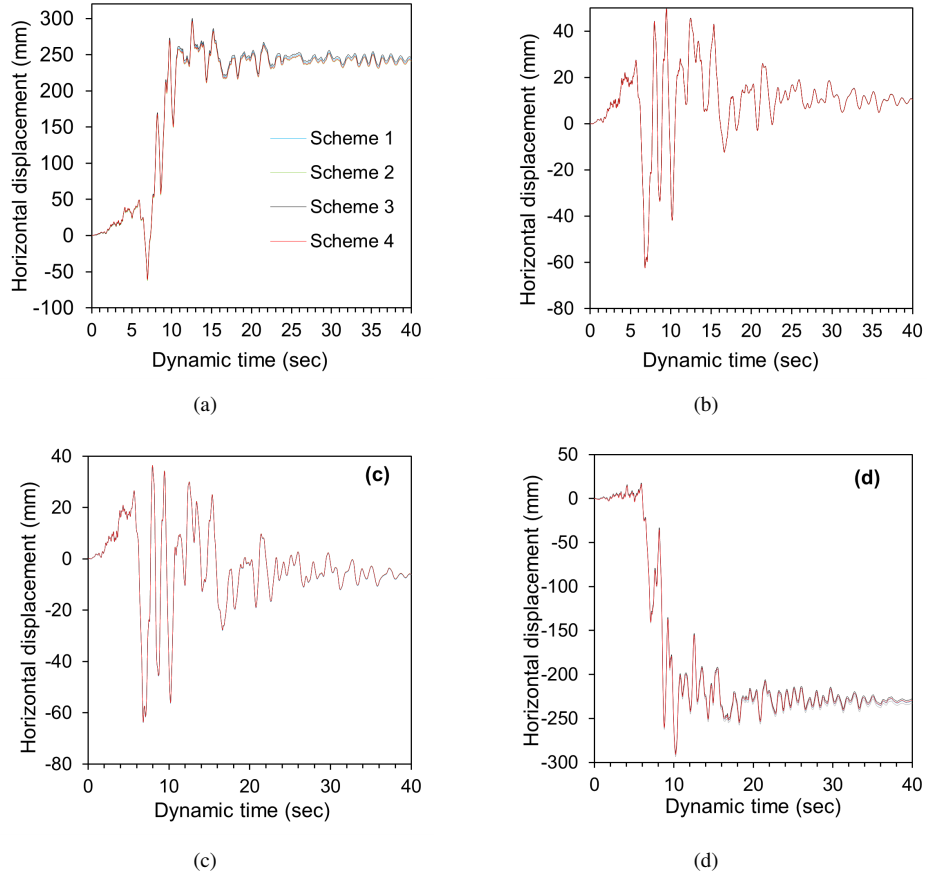


Figure 5. Horizontal displacements of the pit wall during the earthquake: (a) top of the left wall; (b) bottom of the left wall; (c) top of the right wall; and (d) bottom of the right wall

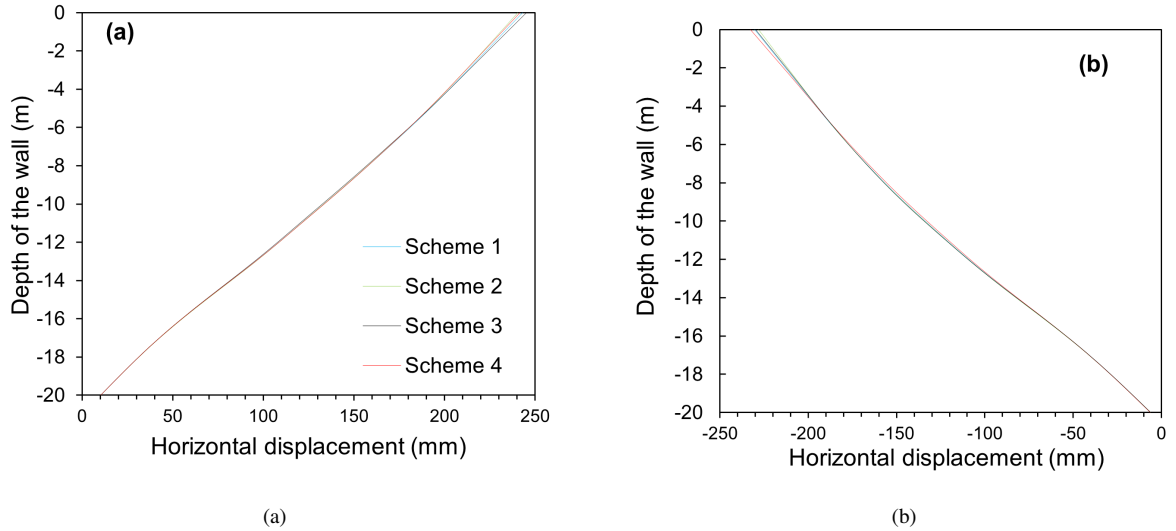


Figure 6. Horizontal displacements of the pit wall after the earthquake: (a) left wall; (b) right wall

3.2 Building Behavior

To fully assess the impact of the prestressing schemes on the excavation pit-soil-building system, the horizontal displacements of each floor of the building located next to a deep excavation pit were analyzed. These data are presented in Figure 7. It is worth noting that, to provide a general idea of the building's behavior during an earthquake,

the analysis was conducted for a scheme with uniform anchor force distribution. Figure 7 shows that the amplitude of the horizontal displacements increases with each floor. Thus, the highest point exhibits the maximum horizontal displacements. Furthermore, this indicates that during an earthquake, the building acts as a cantilever supported from below.

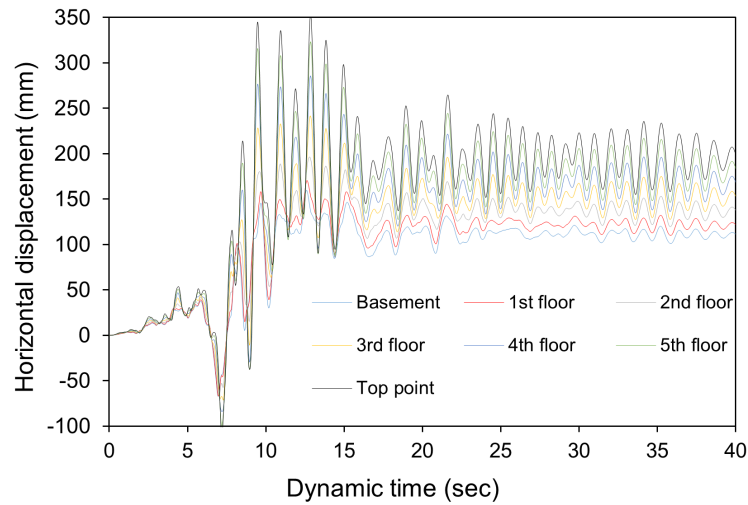


Figure 7. Horizontal displacements of the building during the earthquake for Scheme 1

During an earthquake, each floor of a building behaves differently. Figure 7 shows that the basement, the part of the building closest to the ground, is subject to less seismic action than other parts of the building. The basement oscillates with a minimal period but a maximum frequency, which is natural for a low-rise building. As the building's height increases, the oscillation period increases, and the upper part of the building sways for longer. This can lead to cracks in the building, increased inertial forces, horizontal displacements of the excavation pit wall, and uneven building settlements.

To assess the effect of prestressing the excavation pit wall anchors on the horizontal displacements of the building, a graph was constructed, as shown in Figure 8.

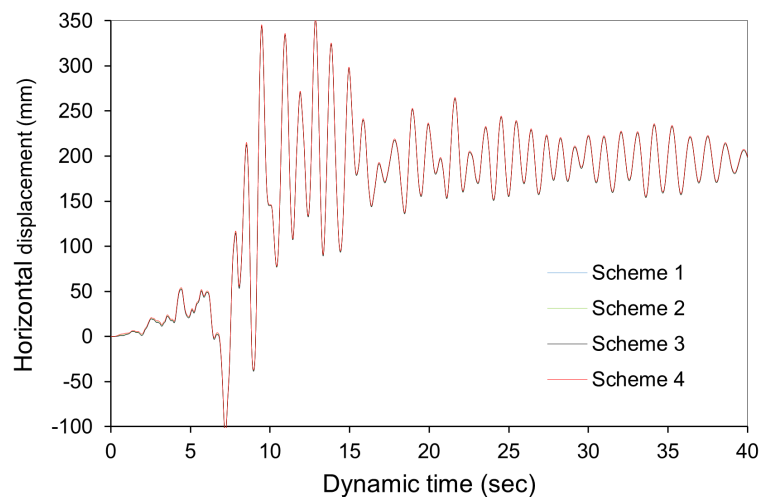


Figure 8. Horizontal displacements of the building's top point during the earthquake

The analysis shows that, with different prestressing schemes, the horizontal displacement curves for the building's top point are virtually identical in shape and amplitude. This indicates that the effect of anchor prestressing on the building's behavior during an earthquake is insignificant. Under dynamic loading, the determining factor is soil deformation, which influences excavation wall displacements and building vibrations. To confirm these findings, an additional comparative analysis of the building's settlements with different prestressing schemes before and after the earthquake was conducted. The results are presented in Figure 9.

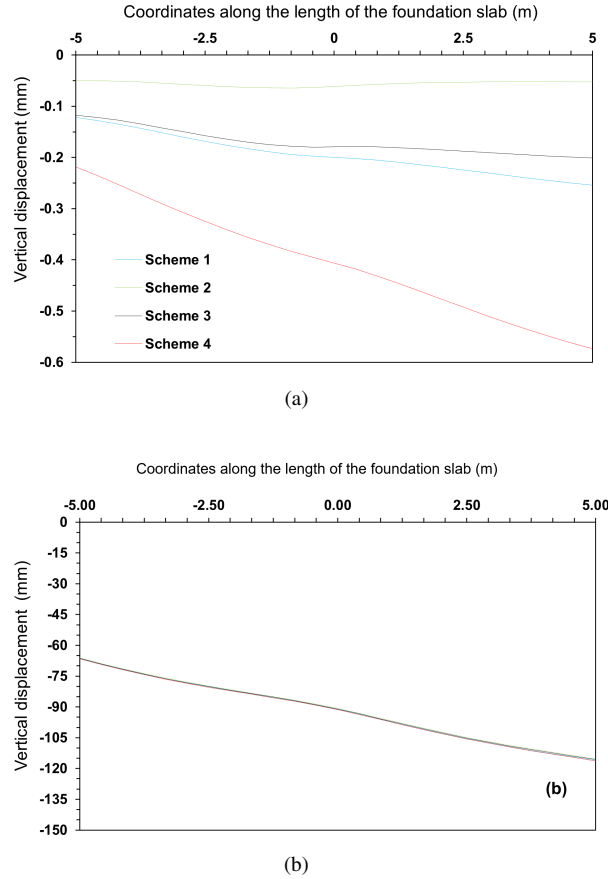


Figure 9. Vertical displacements of the building's foundation: (a) before the earthquake; (b) after the earthquake

Before an earthquake, the most uniform settlement of the building is observed in Scheme 2 with a decreasing prestress pattern. This is explained by the fact that with this pattern, the upper anchors have maximum prestress, which limits soil movement in the most vulnerable, deformation-prone upper part. In all other cases, uneven settlement is observed, which is especially pronounced in Scheme 4. In Scheme 4, where maximum prestress is at the middle anchor level, a local zone of increased wall rigidity appears, limiting wall deformation in this area but leading to increased displacement in the upper and lower parts of the wall. The soil is distributed unevenly, creating zones of unloading, where the wall slopes toward the excavation pit, and zones of overload, where deformation is limited. This results in maximum differential settlement (0.4 mm) compared to other cases.

When comparing the building settlements under uniform and increasing prestressing schemes, it can be observed that the building settles more under the uniform prestressing scheme. This is explained by the fact that in Scheme 1, the lower anchors are prestressed in the same way as the upper ones, even though the soil pressure in this layer is much greater than that in the upper layers. In Scheme 3, the lower part of the wall has greater rigidity, which limits soil deformations and leads to less differential settlement of the building. After an earthquake, building settlements increase sharply several times. This is due to the fact that soil rigidity decreases during an earthquake. Ground vibrations can lead to localized loosening of the soil. Subsequently, the soil becomes more susceptible to deformation. The difference in settlements between opposite sides of the building is 50 mm, indicating the development of differential settlement. This can lead to the building leaning to one side and the development of cracks.

Furthermore, it can be seen that the side closest to the excavation pit settles more, as it is closest to the zone of reduced soil stiffness (the excavation pit area). The building's settlement curves for all prestressing schemes are identical, further demonstrating that soil deformations, which affect every element of the system, are the key factor during an earthquake. As a summary, Table 6 compares the maximum response of the control parameters obtained from the numerical analysis. These parameters include pre-earthquake axial anchor force, post-earthquake axial anchor force, pre-earthquake horizontal displacement of the pit wall, post-earthquake horizontal displacement of the pit wall, pre-earthquake differential settlement of the building's foundation, and post-earthquake differential settlement of the building's foundation. According to these comparisons, the critical schemes are Scheme 3 and 4, where the maximum anchor forces and structural displacements are recorded.

Table 6. Summary of the maximum response of the control parameters obtained from the numerical analysis

Control Parameter	Scheme 1	Scheme 2	Scheme 3	Scheme 4
Pre-earthquake axial anchor force	399.5 kN	459.1 kN	459.4 kN	478.6 kN
Post-earthquake axial anchor force	841.3 kN	808.6 kN	865 kN	825.2 kN
Pre-earthquake horizontal displacement of the pit wall	1.12 mm	1.12 mm	1.52 mm	1.38 mm
Post-earthquake horizontal displacement of the pit wall	243 mm	241 mm	244 mm	242 mm
Pre-earthquake differential settlement of the building's foundation	0.13 mm	0.02 mm	0.08 mm	0.35 mm
Post-earthquake differential settlement of the building's foundation	49 mm	49 mm	49 mm	49 mm

4 Conclusions

An analysis of the impact of different anchor prestressing schemes on the excavation pit-soil-building system showed that the prestressing schemes only affect the system in a static state. In a seismic state, the deformations of the system elements are determined largely by the soil deformations and their impact on the surrounding environment. The initial prestressing values have different effects on the excavation pit walls. For example, with a descending prestressing scheme (Scheme 2), minimal displacements are observed in the upper part of the wall, while with Scheme 4, maximum displacements are observed in the same area of the wall. The deformations of the excavation pit wall in a static system directly depend on the specified initial prestressing values of the anchors at each level. Anchor forces increase by approximately 1.5–2 times after an earthquake for all prestressing schemes. However, despite this, analysis showed that the decreasing prestressing scheme is the most favorable of all the options considered. Scheme 2 ensures a more uniform distribution of forces under both static and seismic conditions, as evidenced by the minimal values of the non-uniformity coefficient. Furthermore, this prestressing scheme results in more uniform building settlement.

Analysis of the study results showed that during an earthquake, the upper part of the wall deforms much more than the lower part. Displacement values increase several times, but the excavation pit walls on the building side experience greater displacements than the walls on the opposite side. This is because the building creates additional load on the excavation pit walls, causing them to shift more. The oscillation period increased with building height. Maximum displacements are observed at the highest point of the building. Upper floors sway longer than lower floors, which can lead to cracks in the building and increased inertial forces. The effect of anchor prestressing on the pit-soil-building system during an earthquake becomes insignificant. The horizontal displacements of the pit wall and the building, as well as its settlement after an earthquake, remain virtually identical regardless of the prestressing scheme.

From a sustainability perspective, an optimal anchor prestressing scheme can improve the safety and efficiency of urban excavations because it reduces material waste and maintenance needs. The results of the present study showed that the decreasing prestressing scheme provides a more uniform force distribution and smaller differential deformations under static and seismic conditions. It can reduce the risk of damage to adjacent buildings and infrastructures. This can minimize repair and reconstruction requirements after earthquakes, reduce material consumption, and improve the efficiency of retaining structures. Therefore, the proposed modeling approach supports sustainable excavation design and enhances the resilience and long-term performance of urban infrastructure systems.

Author Contributions

Conceptualization, M.S.; methodology, M.S.; software, S.A.; validation, S.A.; formal analysis, S.A.; writing—original draft preparation, S.A.; writing—review and editing, M.S.; supervision, M.S. All authors have read and agreed to the published version of the manuscript.

Data Availability

Not applicable.

Conflicts of Interest

The authors declare no conflicts of interest.

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